MODULE 5 NEURONAL INTEGRATION

# Gross Anatomy & Neuronal Integration.

GROSS ANATOMY AND NEURONAL INTEGRATION

Covers gray and white matter, the names for bundles of fibers and clusters of cell bodies, the reflex arc, types of reflexes, and neuronal pools.

BY THE END

1. Distinguish gray matter from white matter.
2. Define the terms tract, nerve, nucleus, and ganglion.
3. Name the five components of a reflex arc in order.
4. Compare the types of reflexes and describe a neuronal pool.

WATCH AFTER YOU READ THIS CHAPTER

## Concept videos: Neuronal integration

drsrennie-stack.github.io/new-build-bio4-solano/nervous-tissue-concept-videos.html

## An Overview

Covers two things: how nervous tissue is organized into the larger structures of the system, and how neurons work together in groups to process information.

THE TERMS THIS PAGE BUILDS ON

| Structure | What it is |
|---|---|
| **Gross organization** | How nervous tissue is arranged into the brain, cord, and nerves |
| **Neuronal integration** | How neurons work together in groups to receive, combine, and respond to information |
| **Gray matter** | Regions made mostly of neuron cell bodies, dendrites, and synapses |
| **White matter** | Regions made mostly of myelinated axons |
| **Reflex** | A fast, automatic, predictable response to a stimulus |

## Gray Matter and White Matter

Nervous tissue sorts into two looks. Where cell bodies gather, the tissue looks gray; where myelinated axons run, it looks white. Compare them.

GRAY MATTER AND WHITE MATTER COMPARED

| Type | What it is made of | Where it is |
|---|---|---|
| **Gray matter** | Neuron cell bodies, dendrites, and unmyelinated fibers | The cortex of the brain and cerebellum, and the inner core of the spinal cord |
| **White matter** | Myelinated axons running in bundles | Beneath the cortex of the brain, and the outer part of the spinal cord |

## Bundles of Fibers and Clusters of Cell Bodies

The same structure has a different name in the CNS and the PNS. Learn the four terms as two pairs: bundles of axons, and clusters of cell bodies.

THE FOUR TERMS, WHERE EACH ONE BELONGS, AND WHAT IT IS

| Term | Where | What it is |
|---|---|---|
| **Tract** | CNS | A bundle of axons within the central nervous system |
| **Nerve** | PNS | A bundle of axons within the peripheral nervous system |
| **Nucleus** | CNS | A cluster of neuron cell bodies within the central nervous system |
| **Ganglion** | PNS | A cluster of neuron cell bodies within the peripheral nervous system |

KEY POINT

Two of the four terms are bundles of axons, tract and nerve, and two are clusters of cell bodies, nucleus and ganglion. Decide which kind of structure you are looking at first, then decide whether you are inside the CNS or outside it.

## The Reflex Arc

A reflex is fast because it runs a short, fixed pathway. Every reflex arc has the same five components, in order.

1. **Receptor** detects the stimulus
2. **Sensory neuron** carries the signal from the receptor to the CNS
3. **Integration center** one or more synapses in the CNS where the signal is processed
4. **Motor neuron** carries the command from the CNS to the effector
5. **Effector** the muscle or gland that carries out the response

## Types of Reflexes

Reflexes are sorted several ways: by the effector, by the number of synapses, and by whether they are inborn. Compare them.

THE TYPES OF REFLEXES AND WHAT EACH NAME IS BASED ON

| Reflex type | Basis of the name | Example |
|---|---|---|
| **Somatic reflex** | The effector is skeletal muscle | The patellar, or knee-jerk, reflex |
| **Autonomic reflex** | The effector is smooth muscle, cardiac muscle, or a gland | The pupillary light reflex |
| **Monosynaptic reflex** | Just one synapse in the integration center | The stretch reflex |
| **Polysynaptic reflex** | Two or more synapses, with interneurons | The withdrawal reflex |
| **Innate reflex** | Present from birth, not learned | The blink reflex |
| **Learned reflex** | Built through practice and repetition | Catching a suddenly dropped object |

## Neuronal Pools and Circuits

Neurons rarely act alone. They are wired into groups, the neuronal pools, and the wiring pattern shapes how a signal is handled.

THE POOLS AND THE CIRCUIT PATTERNS

| Structure | What it is |
|---|---|
| **Neuronal pool** | A functional group of interconnected neurons in the CNS that processes information together |
| **Convergence** | A circuit in which many neurons feed their signals onto one neuron |
| **Divergence** | A circuit in which one neuron spreads its signal to many neurons |
| **Serial processing** | A signal passing along a single straight chain of neurons, as in a reflex arc |
| **Parallel processing** | A signal sent along several pathways at the same time |
| **Integration** | The combining of all the incoming signals so that a neuron produces a single response |

## Sensory and Motor Pathways

Information travels the CNS on long tracts: sensory signals climb upward, motor commands descend, and most pathways cross to the opposite side.

RECEPTORS, PATHWAYS, AND CROSSING

| Structure | What it is |
|---|---|
| **Sensory receptor** | A structure that detects a stimulus and triggers a signal in a sensory neuron |
| **Types of receptors** | Named for what they detect: mechanoreceptors, thermoreceptors, photoreceptors, chemoreceptors, and nociceptors for pain |
| **Ascending pathways** | The tracts that carry sensory information up the spinal cord to the brain |
| **Descending pathways** | The tracts that carry motor commands down from the brain to the body |
| **Decussation** | The crossing of a pathway to the opposite side, so each half of the brain serves the opposite half of the body |

READING THE WIRING IN THE CLINIC

A tap below the kneecap triggers the **patellar reflex**, and a clinician watches the leg kick. That single test checks the whole reflex arc: the receptor, the sensory and motor neurons, and the spinal cord in between. A missing reflex points to a break somewhere along that pathway.

Because most pathways **decussate**, crossing to the opposite side, the right half of the brain controls and senses the left half of the body. This is why a stroke on one side of the brain causes weakness on the opposite side.

The split into **gray** and **white matter** shows up on a brain or spinal cord scan, and a change in that pattern, gray where white belongs or the reverse, is one of the things a radiologist looks for.